

ABHILASH MAKODE

☎ +91- 8767543039 | ✉ abhilash.makode@outlook.com | <https://shorturl.at/ahIH6>

B.E. in Information Technology from Rajiv Gandhi College of Engineering & Research, Nagpur [2011 - 2015 / 61%]

EXPERIENCE

MINDTRAIL TECHNOLOGY PVT. LTD. | Product Based Startup

Jan 2021 – Present

AWS Cloud Engineer/DevOps

Pune

- **Mindtrail Technologies Pvt. Ltd.** is a startup that Provides Products & Services for Small Startups from Development to Deployment
- Extensive experience in defining IT Roadmap, Cloud strategy, Enterprise/Solution architecture, Architecture assessment
- Understand business challenges, problem statements, requirements and existing environments for various clients
- Collaborating with **Developers** and coordinating with **Product** and **Tech teams**, overseeing releases and Deployment on **AWS**
- Architect and Deployed the Infrastructure for the Product. Handled and Troubleshooting **Production Servers, Staging Servers and Development Servers**
- Configured and managed **30+ EC2 instances** across **5 Environments** for the deployment of **Nodejs** and **Reactjs** applications
- Accomplished **Building CICD pipeline** using **GitLab & Jenkins** for **Auto Deployment** of **Artefacts** on Servers
- Composed a **Postgres** read replica set in production along with **automated backups** and restores using **Lifecycle Manager & Bash Scripts**
- Designed, configured, and managed **Nginx-based load balancer** to optimise application performance, ensure high availability through **failover mechanisms**, and implement SSL offloading, resulting in enhanced user experience and efficient resource utilization
- Demonstrated the setup of **Prometheus** and **Grafana** for monitoring the **EC2** machine, **Apache processes** and **Ports**
- Played a crucial part in facilitating the attainment of **SOC 2, ISO, & HIPAA** certifications. Collaborated across teams to implement security controls, evaluate risks, and guarantee alignment with industry standards, resulting in successful audits and certifications
- Successfully **migrated GCP** workloads on-premises. Led assessment, planning, secure migration, security setup, monitoring, recovery, documentation, and post-migration support. Improved resources, data integrity, and alignment with goals
- Devised and executed a robust **Business Continuity Plan (BCP)** for uninterrupted operations during disruptions. Established strategies, processes, and protocols to minimise downtime. Conducted quarterly BCP and **Disaster Recovery drills** to maintain readiness.
- Managed domain configurations, DNS records, SSL/TLS certificates, and security measures on **Cloudflare platform** to optimise performance, enhance security, and ensure seamless website operation
- Implemented robust **AWS patch management** strategies, ensuring timely and secure application of software updates across cloud infrastructure

AWS CLOUD / DEVOPS OPERATIONS | Product Based Start-ups

May 2022 – Present

AWS Cloud Engineer [Freelancer]

Remote

- Worked for multiple startup companies which developed Projects on Education, E-commerce and Fintech Sectors which mostly get developed by small Developers Teams using **AngularJs, ReactJs** and **NodeJs** with **Postgres, MySQL** and **MongoDB** Databases and Deployed on **AWS Cloud**
- Collaborated with teams to design and develop technically compliant, cloud-native enterprise architectures to meet business outcomes

- Performed technical and cost feasibility study, requirement gathering and analysis for Development of solutions that align with business and security requirements
- Responsible for maintaining and Troubleshooting **Disaster Recovery, Fault-Tolerant and Highly Available Infra**
- Administered application and server level monitoring and automatic notifications by using a combination of **AWS Cloud agent, pm2 logs, and AWS Lambda, Nginx**. Monitor **CPU usage, RAM, Volume, Status Failed and 5xx API errors**
- Making Secure Infrastructure from Setting up **SSL** to Securing **Ports** by Referring **ISO, Soc2 and HIPPA Compliance** Strategies
- Accountable for **Building CICD** using **Jenkins** for Autodeploy of **Artefacts** on Servers using **freestyle jobs**
- Used **IAM** for Creating **Users, Groups and Roles** and also implemented **MFA** to provide additional Security to **AWS** Account and its resources
- Creating the configuration for establishing a **VPN tunnel** between on-premise network and **AWS VPC**
- Established **VPC peering** between different **VPCS** for communication between multiple **VPC** resources

PROJECTS

AI Powered CRM Tool | EC2, VPC, S3, SNS, SES, IAM, Cloudflare, Cloudwatch, SSM, System Manager, CloudTrail, ClamAV, Postgres, PgBouncer, Bash script, Jenkins, Bit Defender, OpenVPN

- An Artificial Intelligence powered easy-to-use tools to get more customers & give them a better experience
- Architected and engineered comprehensive infrastructure solutions from the ground up, encompassing design, implementation, and optimization. Delivered scalable, secure, and high-performing systems aligned with business requirements and best practices.
- Planned, built and configured infrastructure within the **VPC** with public and private subnets and configured **routing tables, NAT, Network ACL, Security Groups** and **Internet gateway**
- Planned and executed **EC2 instances**, adapting computing resources for diverse workloads, while implementing **Nginx for load balancing** to ensure efficient traffic distribution in compliance with industry standards
- Effectively managed a cluster of 22 **PostgreSQL databases**, overseeing seamless Prisma migrations and orchestrating deployments to maintain data integrity and application functionality
- Pioneered a **Jenkins-driven multi-stack CI/CD strategy**, optimizing release cycles for different application stacks and elevating operational efficiency and code reliability
- Engineered versatile **Bash scripts** automating tasks like **cross-region snapshot copying, database dump and restore, Prisma migration, .env changes, Git updates, Node crash recovery, and automated environment shutdown and restart**. Resulted in improved efficiency and reduced downtime
- Managed compliance initiatives for **ISO, SOC 2, and HIPAA**, spearheading **Vulnerability Assessment and Penetration Testing (VAPT)** efforts. Ensured alignment with rigorous security standards and regulatory requirements
- Performed quarterly **internal gap assessments** and **executed Business Continuity Planning (BCP)** processes, ensuring continuous alignment with standards and rapid response to potential disruptions
- Set up **CloudWatch Dashboards** to enable real-time infrastructure monitoring, tracking resource utilization, performance metrics, and application health
- Implemented resource rightsizing through custom Bash scripts, coupled with automated environment shutdown during off-peak hours, resulting in noteworthy cost optimization by reducing operational expenses
- Elevated endpoint security by integrating third-party software for access control, encryption, and vulnerability assessments, fortifying defences against cybersecurity risks and safeguarding valuable data
- Led **compliance**-focused meetings, overseeing change, asset, and incident registers, as well as deployment sheets. Ensured alignment with regulatory standards, streamlined processes, and effective incident response within the organisation
- Implemented **Pgbouncer** to efficiently manage PostgreSQL connections, optimizing database performance and resource utilization

Level 3 AI Chatbot | EC2, VPC, S3, ALB, ASG, MongoDB, IAM, ACM, SSM, Route53, RabbitMQ, Prometheus, Grafana & Gitlab, Redis

- **Human-Level Conversational AI** Exclusively fabricated for **E-commerce**. Automates customer interactions on Live Chat, WhatsApp, Instagram, and Messenger
- Level-3 AI helps build store-like experiences on chat, catapulting brands into E-commerce globally.
- 10+ chatbot workflows to make personalised product recommendations and answer common queries
- Designed, constructed, managed, and operated a **scalable, highly available, and fault-tolerant system** from scratch on **AWS**
- Architected custom **VPCs** with Multiple **Public** and **Private Subnets** as per requirements and distributed them as groups into various **AZs**
- Configured **EC2** servers for web application requirements including **Auto Scaling** and **ELB** to ensure high availability
- Implemented **RabbitMQ** for driving toward better user interactions with our applications
- Facilitated **Continuous Integration (CI)** and **Continuous Delivery (CD)** processes
- Creating and maintaining specific **documentation** related to the working environment
- Configured Redis as an in-memory data store, enhancing system performance and enabling fast data retrieval for improved application responsiveness

Online Hobby Class Platform | EC2, VPC, S3, ALB, ASG, KMS, IAM, Route53, WAF, RDS, OpenVPN, SNS, Cloudwatch, Jenkins & GIT, FTP

- An **Online Educational platform** that aims at providing highly engaging, informative and fun classes for children under the age of 12
- Provides extracurricular and co-curricular classes directed by professional mentors to teach prime life skills to little ones
- Planned, built and configured infrastructure within the **VPC** with **public and private subnets** and configured **routing tables, NAT, Network ACL, Security Groups and Internet gateway**
- Created **Snapshots** to take backups of the volumes and images to store **Launch configurations** of the **EC2** instances
- Streamlined **ReactJS** and **NodeJS** web applications using **Nginx** and **PM2**
- Corrected, modified and upgraded **Certbot** to improve **SSL** connection and setup automated renewal
- Configured and Maintained the **Jenkins build** and **Automation jobs**
- formulated Amazon **Route53** to manage **DNS** by creating and managing **Hosted zones** and **Recordsets**
- Implemented **FTP, SFTP & AWS Transfer Family Service** to facilitate seamless file transfers, ensuring efficient sharing and management of data across networked systems

OTHER EXPERIENCES

Hexaware Technologies Limited

Support Engineer

March 2018 – July 2019

Nagpur

Online Educational platform

Designed the **E-Learning** courses for the client which teaches various policies which are required to work in a company environment

City Service Provider

Online classifieds platform to empower every person in the City to independently connect with buyers and sellers online and make secure transactions.

Job Consultant Software

Web-based online service implemented to create one healthy environment, where Employers, Consultants, Institutions and Jobseeker can share their ideas and post and search jobs.

TECHNICAL SKILLS

AWS Services: EC2, VPC, S3, Route53, CloudWatch, SNS, (ASG)Autoscaling, ELB, EBS, IAM, SSM, KMS, SES, WAF, ACM(Certificate Manager), RDS, Lambda, AWS CLI, CloudTrail, System Manager

DevOps: Git, Docker, Jenkins, Prometheus, Grafana

Operating systems: Windows, Ubuntu

WebServers: Nginx, Apache

Scripting Language: Bash, Shell

Databases: Postgres, MongoDB, MySQL

Security: BitDefender Endpoint Security, ClamAV

Abhilash Makode

Date :